[Metric]
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FEDERAL SPECIFICATION

CHAIRS, ROTARY AND STRAIGHT, (ALUMINUM, OFFICE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rotary and straight office chairs that are padded and upholstered with an exposed aluminum frame.

1.2 Classification.

1.2.1 Types, classes, and styles. The rotary and straight chairs covered by this specification shall be of the following types, classes, and styles, as specified (see 6.1):

Type I - Straight (leg) chairs.

Class 1 - General use.

Style A - With arms.
Style B - Without arms.

Type II - Rotary (swivel) chairs.

Class 1 - Tilting seat and tilting adjustable back.

Style A - With arms.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

FSC 7110

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Federal specifications:

FF-C-82	- Casters, Swivel, Chair.
QQ-A-200/8	- Aluminum Alloy Bar, Rod, Shapes, Tube, Wire, Extruded and Structural Shapes, 6061.
QQ-A-250/11	- Aluminum Alloy 6061 Plate and Sheet.
QQ-A-591	- Aluminum Alloy Die Castings.
QQ-A-596	- Aluminum Alloy Permanent and Semi-Permanent Mold Castings.
QQ-A-601	- Aluminum Alloy Sand Castings.
WW-T-700/6	- Tube, Aluminum Alloy, Drawn, Seamless, 6061.
CCC-A-680	- Artificial Leather (Cloth Coated), Vinyl Resin, Expanded Layer (Upholstery).
CCC-C-700	- Cloth, Coated, Vinyl Coated (Artificial Leather).

Federal Standard:

Fed. Std. No. 245 - Tolerances for Aluminum Alloy and Magnesium Alloy Wrought Products.

(Activities outside the Federal Government may obtain Federal Specifications and Standards required for bidding purposes from Business Services Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, and Auburn, WA.

Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Standards required by contractors in connection with specific procurement functions can be obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standard:

ANSI/BIFMA X5.1-1985 - General-Purpose Office Chairs - Tests.

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(Application for copies may be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

American Society for Testing and Materials Standards:

ASTM B 110 - Dielectric Strength of Anodically Coated Aluminum.
ASTM B 136 - Stain Resistance of Anodic Coatings on Aluminum.
ASTM B 137 - Test for Weight of Coating on Anodically Coated Aluminum.
ASTM D 3453 - Flexible Cellular Materials - Urethane for Furniture and Automotive Cushioning, Bedding and Similar Applications.
ASTM D 3574 - Flexible Cellular Materials - Slab, Bonded, and Molded Urethane Foams.
ASTM D 3597 - Woven Upholstery Fabrics - Plain, Tufted or Flocked.
ASTM D 3770 - Flexible Cellular Materials - High-Resilience Polyurethane Foam (HR).

(Application for copies may be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Fire Protection Association Standard:

NFPA 261 - Resistance of Mock-Up Upholstered Furniture Material Assemblies to Ignition by Smoldering Cigarettes.

(Application for copies may be addressed to the National Fire Protection Association, Inc., Batterymarch Park, Quincy, MA 02269.)

State of California Standard:

Technical Information Bulletin 117 - Requirements, Test Procedure and Apparatus for Testing the Flame Retardance of Resilient Filling Materials used in Upholstered Furniture.

(Application for copies should be addressed to the State of California. Department of Consumer Affairs, Bureau of Home Furnishings, North Highlands, CA 95660.)

3. REQUIREMENTS

3.1 Materials. All materials used shall be as specified herein and shall meet the quality standards established in the respective material paragraphs. Any material required which is not named or which is named but for which a standard of quality is not specified shall be of the quality used for standard commercial items of the kind covered by this specification

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Materials shall not have been previously used in their present state. They shall be free of defects which affect the appearance or the serviceability of the finished product.

3.1.1 Aluminum tube. Aluminum tube used in the fabrication of chairs shall be of the dimensions specified herein and, except as specified, shall conform to either WW-T-700/6, alloy 6061 "0" temper, or QQ-A-200/8, alloy 6061 "0" (annealed) temper. When specified herein (see 3.2.2) tube may be T4 (solution heat-treated) or T6 (solution heat-treated and then artificially aged) temper in lieu of "0" temper.

3.1.2 Aluminum plates and sheets. Aluminum plates and sheets used in the fabrication of the chairs shall be of the dimensions specified herein and, except as specified, shall conform to the requirements of QQ-A-250/11, "0" (annealed) temper. When specified herein (see 3.2.2) temper may be T4 (solution heat-treated) or T6 (solution heat-treated and then artificially aged) of QQ-A-250/11.

3.1.3 Aluminum casting. When an aluminum casting is furnished as the outer shell of the swivel base, it shall conform to one of the following: alloy 142 and T61 temper; alloy B195 and T6 temper; Tenzaloy and T5 temper of QQ-A-596; alloy 380 of QQ-A-591; alloy 384 and F temper. If sand casting is used, the alloy shall be Almag 35 and T4 temper of QQ-A-601. When castings are used as component parts of a wrought aluminum swivel base, they shall be cast of class 8 (alloy 356) of QQ-A-596.

3.1.4 Steel parts. All steel parts shall be cut, stamped or die-formed from steel sheets or steel tubing. The steel shall be bright, smooth, free of scale, pits, wrinkles, laps, crimps, and buckles affecting strength.

3.1.5 Finish coating. The finish coating for fabricated aluminum or cast aluminum parts shall be either a lacquer, a varnish, or an enamel. Application and curing shall be as recommended by the manufacturer. The coating shall produce a uniform and tough finish with good adhesion to aluminum and without runs, wrinkles, orange peel or other surface defects.

3.1.6 Polyurethane foam. The polyurethane foam shall be Grade HR-I for the back cushion and Grade HR-II for the seat cushion, slab or molded high-resilience foam complying with all requirements of ASTM D 3770. Alternatively, the polyurethane foam shall not contain any filler material and shall be molded or slab complying with all requirements of ASTM D 3453 for the grades designated in Table I below; the density shown in Table I below shall be tested in accordance with ASTM D 3574. All polyurethane foam shall comply with the flammability requirements of NFPA 261 and State of California Technical Information Bulletin 117.

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TABLE I. Requirements of polyurethane foam.

	Back Cushion	Seat Cushion
Grade	34	44
Dynamic Performance Grade	BD	AD
Static Performance Grade	BS	AS
Density (kg/cu. m)	32-40	43-56

3.1.7 Wadding/thin fiber batting. The wadding/thin fiber batting shall be white cotton or synthetic fibers, 4.5 mm to 14.5 mm thick, smoothly carded, thermally bonded, or resin treated. The density shall be 67 g/cu. m to 167 g/cu. m. It shall comply with the flammability requirements of NFPA 261 and California Technical Information Bulletin 117.

3.1.8 Artificial leather. The artificial leather shall be either Class 2 or better, Treatment a, conforming to CCC-C-700 or shall be Class 2, Treatment (a) 1, conforming to CCC-A-680. The finish or grain shall simulate top-grain leather. Unless otherwise specified (see 6.1), the color shall closely match the following Government standard sample as specified in the invitation for bids or request for proposal:

Black FSS-V-01008

Government standard samples are available from the General Services Administration, Federal Supply Service, Furniture Commodity Center (3FNE-CO), Washington, DC 20406.

3.1.9 Upholstery fabric. The upholstery fabric shall be 100% flat nylon, dyed, plain weave, not less than 650 g/linear m (based on 1372 + 2 mm width) exclusive of back coating. The fabric shall comply with all requirements of ASTM D 3597, NFPA 261, State of California Technical Information Bulletin 117 and with the following deviations from ASTM D 3597:

A) In section 7.6, delete Note 6 entirely.

B) Delete sections 7.6.2, 7.6.3, and 7.6.4 entirely.

C) In Table I, change requirements of:

Breaking strength (load) to 1110 N, min. for warp and filling.
Tongue tear strength to 90 N, min. for warp and filling.
Resistance to yarn slippage to 180 N, min. for warp and filling.

D) In Table I, Surface abrasion shall pass heavy-duty requirements.

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Unless otherwise specified (see 6.1), the color shall closely match the following Government standard samples as specified in the invitation for bids or request for proposal:

Black	FSS-F-01009
Black/White	FSS-F-01014
Red/Black	FSS-F-01013
Green/Black	FSS-F-01016

Government standard samples are available from the General Services Administration, Federal Supply Service, Furniture Commodity Center (3FNE-CO), Washington, DC 20406.

3.1.10 Glides. The glides shall be properly nickel or chromium plated, or polished stainless steel and shall be of the live-rubber-cushioned and spring steel expansion, rubber expansion, PVC cushion or nylon cushion type. The glides shall be firmly secured to each leg but shall be capable of removal without the use of special tools. There shall be no metal-to-metal contact of glide base and chair leg after the glides are applied. The glides shall have a diameter of not less than 23 mm nor more than 29 mm. The glides shall have a substantially flat bottom. The base of the glide shall be stamped from not less than 1.2 mm thick steel and shall be case hardened after forming to a Rockwell hardness of not less than 15 N 85 or equivalent on a comparable scale. The base shall have properly curled edges to contain and retain the cushion which shall fill the inside of the base and extend not less than 3 mm above it.

3.1.11 Casters. The casters shall be Type I, Style A or C of FF-C-82. The pintles of stem type casters shall be not less than 9.5 mm diameter. The wheel tread material shall be rubber or plastic with a hardness of 35 to 50 at 21 ± 1 degrees C as measured on the "D" scale of the Shore durometer. The wheel diameter shall be 63.5 ± 1.6 mm.

3.1.12 Sound deadening material. The sound deadening material for use in each of the four legs of the wrought or die formed bases shall be composed of not less than 78 kg test, corrugated fiber board. The fiber board shall be scored and creased from die cut sheets to form not less than three expanding folds so that the edges will be in firm contact with the inside surface of the base leg.

3.1.13 Extruded bumpers. The internal bumper on back of chairs shall be extruded angle shaped vinyl or rubber, 127 ± 2 mm long and 1.6 ± 0.4 mm thick, with each leg 19 ± 1 mm wide. These shall be firmly applied to each side of back frame without adhesives or screws and located with the upper edge 22 ± 1 mm below the top of back frame.

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3.2 Design and general construction.

3.2.1 Design. The general exterior appearance of the chairs covered by this specification shall be as shown in figures 1, 2, and 3. The illustrations depicted identify the basic styling required but do not represent specific requirements unless so stated herein.

3.2.2 Assembly of aluminum parts. Aluminum alloy exterior parts shall be formed from 0 temper material in accordance with the measurements specified herein. Chairs shall be assembled by the inert-gas shielded-metal-arc process. All unfused flux shall be completely removed and all exposed welds shall be ground smooth where necessary to produce a finished appearance. Jigs shall be used to insure proper alignment of parts during welding. Mechanical joints shall be made with white metal plated steel bolts, nuts, and lock washers, and shall be reinforced by metal inserts. After the aluminum assemblies are formed, welded, and cleaned, the entire units shall be heat treated. They shall then be immediately moved into an air chamber with air circulated around the assemblies or water quenched so that the temperature of the assemblies is reduced to room temperature. Alternatively, chairs without welded structural joints may be formed from T-4 or T-6 temper. Chairs formed from 0 temper or T4 temper shall be subsequently aged to T6 temper.

3.2.3 Forming parts. The tubes, when bent, shall have true and even curves without flattening, crushing or crimping that will affect the appearance or strength of the tubes. Grinding which reduces thickness below 75 percent of the original will not be permitted. All bends in both sheets and tubing shall be free from fracture. Any fracture occurring in manufacture shall be discarded. Rewelding of such defects will not be acceptable.

3.2.4 Application of upholstery. The upholstery covers shall be securely attached to the seat and back pans by adhesives and bayonets or by spring metal clips. Upholstery shall be applied in a manner to assure smooth, even, well rounded edges that will be free from hazardous surfaces after continued heavy use. All seams and welts shall present a neat appearance with true even lines.

3.2.5 Chair controls. The Type II chair shall be fitted with a mechanism for adjusting the tension against tilting of the seat. The adjustment for tilting and raising the seat shall be made by means of metal handles or hand wheels. An adjustment shall be provided for raising the seat over a range of not less than 76 mm as shown in figure 3. The adjustments shall be self-locking and shall permit revolving without changing the height of the seat. The chair shall be fitted with a mechanism for adjusting the

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tension against tilting of the back rest by the use of springs or rubber torsion units. The adjustments for horizontal movement of the back rest, raising the seat not less than 76 mm, and raising the back rest not less than 12 mm shall be made by means of metal handles or hand wheels. The horizontal and vertical adjustment of the back rest shall be easily adjusted and shall be rigid without wobble in the mechanism.

3.2.5.1 Chassis, spider-arms and spindle. The spindle shall be cold drawn steel 25.40 + 0.00, -0.05 mm in diameter except the portion that shall be welded or pressed fit to the chassis shall be not less than 19.0 mm in diameter. Either cut thread or rolled thread spindles may be used. The threads on the cut thread spindle shall terminate at a minimum of 63.5 mm from the bottom end. On a rolled thread spindle the threads may extend full length but the keyway slot shall terminate at a minimum of 63.5 mm from the bottom end of the spindle and a steel J-washer shall be used. When the top of the spindle housing (base hub) does not have a plastic bearing surface, a replaceable plastic J-washer shall be inserted on the rolled thread spindle between the steel J-washer and the top of the spindle housing. The threads and the keyway slot on all spindles shall be free of burrs and sharp edges. An iron casting of proper shape shall form the chassis into which the spindle shall be pressed fit for a distance of not less than 49.2 mm. Through the casting and spindle a 4.8 $\pm$ 0.1 mm diameter rivet shall be driven. When a casting is not used as the chassis, the spindle shall be further reinforced by a bracket, tube or an accurately machined part extending around and along the spindle not less than 19.0 mm to provide additional strength. The spider arm shall be formed to provide a vertical web of not less than 25.4 mm and shall be designed for rigid fastening to the seat frame structure at not less than four places with bolts, lock washers and nuts or hex head self-threading screws.

3.2.5.2 Tilting unit. Tilting units shall provide the tension against the tilting of the chair seat and back by means of steel springs or rubber torsion units. Tilting tension bolts shall be not less than 8.73 mm diameter, and must be entirely supported by the chair control structure when a wrought or die formed base is used. The bolt shall have ample clearance through the entire tilting range to prevent breaking due to wear by friction. The tilting unit shall have a positive safety stop that will limit the tilting action during normal use and in case of failure or breakage of the tension unit. The range of the tilting action shall be not more than 0.42 rad.

3.3 Detail construction. The construction requirements specified herein are minimum. The contractor shall exceed these requirements (without altering the general configuration, basic

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concept or primary construction specified or illustrated herein) wherever necessary to conform to the test provisions specified herein. Any detail not specified herein shall be consistent with good quality production of similar commercial products. All elements of construction shall be free of any conditions which hinder the proper function of any part of the units or which constitute a potential hazard to personnel or their clothing.

3.3.1 Type I, Class 1, Style A. The type I, class 1, style A chair shall be of the general design and minimum dimensions (except as otherwise specified) shown in figure 1 and as specified herein.

3.3.1.1 Frame. The components of the chair frame shall be assembled as specified in 3.3.2. The lower end of all legs shall be neatly closed by leg blocks and shall be pierced to receive and secure the glides specified in 3.1.10. The lower ends of the arm rest tubing shall be closed by notching and welding into a neat curve where it is secured to the seat frame.

3.3.1.2 Seat pan and pad. The seat pan shall be saddle shaped, perforated with sufficient holes (approximately 30, about 6 mm diameter) for proper air circulation and be not less than 0.91 mm thick steel. The front and side of the seat pan shall be formed downward at right angles to the top of the seat pan and shall overlap the front frame for not less than 15 mm and the sides up to the arm posts or back frame. A bumper as specified in 3.1.13 shall be applied to the edge of the seat pan which is flanged downward. The seat pan shall be padded with not less than 25 mm thick polyurethane foam slab conforming to 3.1.6, and covered with the upholstery cover material specified in 3.1.8. The upholstery covers shall be applied as specified in 3.2.4. The seat pan shall be secured to the frame with not less than four bolts and clips applied to the underside of the seat frame.

3.3.1.3 Covered back. The back pan shall be of not less than 0.75 mm thick steel and shall conform to the top frame and the back brace in both shape and contour. The edges of the pan shall be hemmed to prevent damage to the upholstery material and the center portion recessed within the back frame and back brace. The pan shall be padded with not less than 12 mm thick polyurethane foam slab conforming to 3.1.6, which shall cover the entire pan and extend slightly over the edge to provide cushioning at the edges of the pan. The upholstery cover shall be as specified in 3.1.8 and shall be applied as specified in 3.2.4. The upholstery cover shall be applied to both front and rear surfaces of the back rest. The back rest shall be secured to the back frame by two screws at the bottom and not less than two clips at the top, or by four concealed screws.

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3.3.1.4 Arm pads. The arm pads shall be of semi-hard rubber or semi-hard, molded plastic. They shall have a Shore durometer hardness of 50 to 70, D scale or a Rockwell hardness of 50 to 70, M scale. The top surface shall be textured with a black stippled or matte finish. The arm pads shall cover the top of the arm and shall be not more than 45 mm wide. Minimum thickness shall be 7 mm as measured from the arm rest surface which rests on the arm support to the top surface of the arm rest. They shall extend around the contour of the arm from a point not to exceed 64 mm from the back of the frame to not less than 50 mm below the top of the arm support. The arm pads shall be secured to the arm frame with a concealed stud and screws or with concealed screws.

3.3.2 Type I, Class 1, Style B. The type I, class 1, style B chair shall be of the general design and minimum dimensions (except as otherwise specified) shown in figure 2 and as specified herein. The chairs shall be as specified for the type I, class 1, style A chairs except that arms shall be omitted.

3.3.3 Type II, Class 1, Style A. The type II, class 1, style A chair shall be of the general design and minimum dimensions (except as otherwise specified) shown in figure 3 and as specified herein. The aluminum parts of the chair shall meet the minimum requirements in the table below:

Part	Dimensions (mm)	Thickness (mm)
Arms	28.6 X 22.2	1.65
Base (if die formed)		1.65
Back frame	28.6 X 28.6	1.65
Back brace	10.3 X 25.4	1.47

3.3.3.1 Seat frame and seat. The seat frame shall be not less than 0.87 mm thick channel-formed steel and shall be reinforced to prevent warping. A diaphragm made of not less than 422 g duck shall be stretched over a separate metal frame and securely fastened thereto with metal stitching. Alternatively, the diaphragm may consist of a covering of not less than 422 g duck stretched over the seat frame. The duck shall be securely fastened to all four sides of the entire vertical face and bottom flange of the seat frame with adhesive, as recommended by the adhesive manufacturer. In addition to the adhesive used to secure the duck, 16 ± 1 mm wide, barbed, metal spring clips spaced on not more than 64 mm centers shall be used to secure both the duck and the seat cover to the bottom return flange of seat frame. The diaphragm shall then be secured to the seat frame and shall be supported by not less than four serpentine shaped steel springs of not less than 3.80 mm diameter spring

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wire, extending from side to side and spaced uniformly. The springs shall be securely fastened to the top flange of the seat frame. The seat cushion shall be polyurethane foam as specified in 3.1.6. The cushion, before upholstery is applied, shall be not less than 50 mm thick at the front of the seat, and not less than 25 mm thick at the back. Arms shall be secured to the seat frame and both ends of the arm rest tubing shall be closed by notching and welding into a neat curve where secured to the seat frame. One layer of wadding as specified in 3.1.7 shall be cemented to all sides of the seat frame. Seat construction shall be covered with a tailored cloth cover conforming to 3.1.9 and applied as specified in 3.2.4. The seat frame shall be reinforced with steel brackets at the point of attachment and secured to the spider at four different places with bolts, lock washers and nuts or hex head, self-threading screws. Machine buffed leather bumpers or bumpers covered with the same cloth as the other upholstery shall be attached to each front corner of the seat.

3.3.3.2 Back frame. The back frame shall be formed from one continuous piece into a "U" shape. One cross brace shall be welded between the legs of the back frame. The lower part of the frame shall have in each leg an aluminum bar reinforcement to which the upper fastenings for the cross member of the chair control, back tilt unit is secured. Aluminum pivot bearing plates shall be assembled into the open ends of the back frame. These pivot plates as well as reinforcements at the bottom shall be made from 9.5 ± 0.4 mm by 23.7 ± 0.4 mm wide extruded aluminum bar. Alternatively, the reinforcements at the bottom may be cast aluminum, the minimum dimensions of which shall be 19.0 mm by 47.6 mm long. The top ends of tubing shall be closed by notching and welding into a graceful radius. The back frame shall be adjustable vertically and horizontally.

3.3.3.3 Cushioned back. The back shall consist of a steel formed angle frame not less than 2.85 mm thick to which a steel pan not less than 0.57 mm thick shall be welded. The back shall be upholstered with a contour shaped, polyurethane foam cushion as specified in 3.1.6, tapering from a thickness of not less than 44 mm or more than 51 mm at the top to not less than 92 mm or more than 98 mm at the bottom. The entire back, including the steel back pan shall be covered with the same matching upholstery material used for the seat by machine sewing the covering into an envelope shape, fitting over the back unit and securing in place by hand sewing across the bottom edge forming a completely enclosed, tailored back unit. The upholstered back shall be assembled to the aluminum back frame with two metal pivot pins which will permit the back unit to respond to movements of the occupant's back. The angle adjustment of the back shall be obtained by a rubber tipped adjustment screw located in the center lower edge of the back which can be locked in place with a

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round lock nut without the use of tools. Adjustment screws shall have end threads upset so that at maximum adjustment they will not become disengaged from the back.

3.3.3.4 Arm pads. The arm pads shall be formed steel channels of not less than 0.87 mm thick steel shaped to fit the contour of the chair arm frames. The formed steel channels shall be padded on the top and sides with polyurethane foam conforming to 3.1.6. The padding shall be a minimum of 19 mm thick securely cementedto the steel channels and covered with a tailored cover secured in place by cementing plus the use of not less than six spring clips in each pad. The upholstery cover shall be as specified in paragraph 3.1.8, the color of which shall harmonize with the color of the upholstery fabric for the seat and back. The arm pads shall be secured to the arm frame with concealed studs and screws or concealed screws.

3.3.3.5 Chair control. The chair control shall be as specified in 3.2.5.

3.3.3.6 Base. The base shall be in accordance with 3.3.3.6.1 or 3.3.3.6.2.

3.3.3.6.1 Type I base. The aluminum base shall be fabricated of materials not less than the thickness specified in the table in 3.3.1. When a cast aluminum base is furnished, it shall be as specified in 3.1.3 and anodized to match the other chair parts as specified in 3.4.1.1. The base shall be reinforced to provide continuous support throughout the vertical depth of the hub. The hub shall consist of one of the following: A seamless steel tube not less than 1.57 mm thick with an outside diameter of not less than 28.5 mm; a welded steel tubing not less than 31.7 mm in outside diameter having a liner with self lubricating qualities, or a commercial type, insertable hub having self lubricating bearings with the lower end of the steel tube flanged inward to hold the bearings in place, or a cast aluminum hub with a self lubricating liner. The seamless steel tubing hub shall be rolled securely and tightly over the top and bottom. The self lubricated liner of the welded steel tubing or cast aluminum hub shall be molded of virgin nylon resin and shall provide a one-piece full length, vertical, rotational bearing support for the spindle and top bearing support for the height adjustment hand-wheel of the chair control. The liner shall have full steel or aluminum backing to provide maximum structural strength. The commercial type insertable hub shall have a collar or flange plate not less than 63.5 mm in diameter for securing to the base with at least four, 6.4 $\pm$ 0.2 mm diameter screws and shall have a smooth center area 4.8 $\pm$ 0.2 mm wide to serve as a slot for the chair control thrust bearing. A 63.5 $\pm$ 1.0 mm lock plate shall

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be secured to the base by a minimum of three screws when the cast aluminum hub is provided. The lock plate shall provide 4.0 ± 0.2 mm support to the surface of the chair control thrust bearing. The inside diameter of the hub shall be accurately sized and burnished to receive and vertically align the spindle of the chair control. The chair control thrust bearing shall be sintered powdered metal or molded nylon; or if a steel hub with a molded nylon top bearing support is used, it shall be of case hardened steel. The base shall have steel caster sockets, or aluminum plates if style C casters of FF-C-82 are used. The sockets or plates shall be securely mounted to retain the casters in a position which will bring the vertical axis of the caster bearing parallel to the chair control spindle. When pintle or stem type casters are furnished in an aluminum cast base, the caster retainer shall be a seamless steel tube molded as an integral steel insert into the aluminum casting. When pintle or stem type casters are furnished in a wrought aluminum base the caster retainer shall be a seamless steel tube molded as an integral steel insert in an aluminum casting. This separate caster socket casting shall then be welded around the entire periphery to the wrought base leg. The wrought base shall be sound deadened as specified in 3.1.12. Provisions shall be made for securing the chair control to the base which will permit revolving and removal of the spindle from the hub when desired.

3.3.3.6.2 Type II base. The cast aluminum base shall be of an alloy as specified in 3.1.3 and shall be anodized to match the other chair parts as specified in 3.4.1.1, except the die cast base shall be properly cleaned in the etchant of the anodized process and rinsed thoroughly. The base legs shall be "T" section design and the center section of the base shall extend downward to provide full support to a pressed fit molded nylon hub with an integral lip which shall provide a flat, smooth, bearing surface not less than 4.7 mm wide. The center hole of the hub shall be of the size to accept the chair control spindle to provide maximum vertical and rotational bearing support. Alternatively, the hub and liner may be in accordance with any permitted in 3.3.1.6.1. The caster sockets shall be reamed to properly hold the caster retainer ring and to provide a secure fit of the caster. Provision shall be made for securing the chair control to the base which will permit revolving and removal of the spindle from the hub when desired. The chair control thrust bearing shall be compatible nylon or case hardened steel.

3.3.3.7 Casters. The casters shall be as specified in 3.1.11.

3.4 Method of finish.

3.4.1 Aluminum parts.

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3.4.1.1 Metal finish and anodic coating. All exposed aluminum parts shall have a satin finish produced by polishing the aluminum surfaces with a suitable abrasive material to obtain a fine directional finish. All exposed aluminum parts (unit bases and frames) shall be of the same grain and of similar finish. After cleaning, aluminum components, excluding those produced from alloy 380, shall be anodized in a sulfuric acid electrolyte. The anodic coating weight shall be not less than 0.024 mg/sq. mm as determined according to ASTM B 137. All anodized components shall be sealed in boiling water to sufficient purity to render the coating impervious to stains when tested in accordance with ASTM B 136. Anodic coating on specimens tested in accordance with ASTM B 110 shall not break down when less than 100 volts has been applied to the specimen. The exposed aluminum surfaces shall be uniform in appearance and shall be free of small pits, flow lines and color differences.

3.4.1.2 Protective coating. Sufficient coats of finish as specified in 3.1.5 shall be applied over the anodic coating to produce a minimum dry film thickness of 0.010 mm. Blemishes, blowholes, crevasses and other imperfections in casting shall not be concealed by welding, putty or other filler, nor shall such defects be covered with the coating.

3.4.2 Steel parts. All steel parts shall be made rust resistant by white metal plating, or by chemical cleaning followed by a finish coating as specified in 3.1.5. The chair control shall be smoothly coated with aluminum paint to closely match the color of the aluminum parts of the chair or with a neutral shade of gray. Alternatively, handwheels, handles, and bolts of chair control may be metal plated.

3.5 Identification marking. The bottom of the chair seat shall be legibly and permanently marked with the following information: Manufacturer's name or trademark, contract or order number, national stock number, and month and year of manufacture.

3.6 Workmanship. Finished chairs shall be free from any defects that affect the appearance, function or serviceability of the chairs, or that could cause personal injury or damage to clothing.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.1), the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right

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to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article sample inspection. Examination and testing of the first article sample (when required) shall be made on completely fabricated items for all provisions of this specification applicable to end product examination and tests.

4.3 Sampling for inspection and acceptance. Sampling for lot inspection and acceptance shall be performed in accordance with MIL-STD-105, except as otherwise specified herein.

4.3.1 Component and material inspection. In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested and inspected in accordance with this specification.

4.3.2 Inspection of end items. The lot shall be all chairs of the same type, style, and class. The lot size shall be expressed in units of chairs for the purpose of determining the sample size in accordance with MIL-STD-105. The sample unit for this examination shall be one completely fabricated chair.

4.3.2.1 Visual examination. The chairs shall be examined for the defects listed in the table below. The inspection levels and the acceptable quality levels shall be as specified in 4.4.

Classification of Defects

Examine	Defect	Major	Minor
Enamel or lacquer finish	Indication of rust under coating.	X	
	Poor adhesion, i.e., blistered, checked, peeling.		X
	Tacky (not dry to touch).		X
	Wrong type or color.		X
	Not smooth or uniform.		X
	Foreign matter in coating.		X
	Scratch through to base metal or bare spots.	X	
	Abrasions or area of thin coating.		X

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Classification of Defects (cont.)

Examine	Defect	Major	Minor
Metallic coatings	Wrong kind.		X
	Not uniform.		X
	Poor adhesion, i.e., blistered, peeling, stained or discolored.		X
Metal parts (material and finish).	Not material specified.	X	
	Aluminum not satin finish.	X	
	Rust, scale or other imperfection.		X
	Finish of exposed aluminum parts not uniform.		X
	Steel parts not painted or plated.	X	
Upholstery material	Not material specified.	X	
	Wrong color.	X	
	Misweave, i.e., multiple float or other fabric damage affecting serviceability.	X	
	Material has objectional odor.		X
	Stained, streaked, spotted or greasy.		X
	Faded, or shade variations affecting appearance.		X
Upholstery (workmanship)	Not securely attached.		X
	Not attached by adhesive and bayonets, (except when adhesive only is permitted) or by spring clips.		X
	Lines of seams and welts not true and even.		X

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Classification of Defects (cont.)

Examine	Defect	Major	Minor
Casters and glides	Missing.	X	
	Not types specified.		X
Arm pads	Not specified material.	X	
Arm pads (cont.)	Fastenings not concealed wnen attached to chair frame.	X	
	Top surface does not harmonize with back and seat upholstery.		X
Base	Not reinforced to provide continuous support through depth of hub.	X	
	Hub not one of types specified.	X	
	Not equipped with steel caster sockets for Style A casters or aluminum plates for Style C casters.	X	
	Wrought base not sound deadened.		X
Seat and back pans	Seat pans not saddle shaped.	X	
	Seat pans not perforated as specified.		X
	Back pans not contour shaped.		X
Chair controls	Seat height adjustments not self locking.	X	
	Seat height changes when chair revolves.	X	
	Spider arm fastened to seat frame structure at less than four places.		X
	Tilting unit not provided with positive safety stop.	X	
	Not fitted with mechanism for		

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Classification of Defects (cont.)

Examine	Defect	Major	Minor
Chair controls (cont.)	adjusting tension against tilting of back rest by use of springs or rubber torsion units.	X	
	Adjustment for horizontal movement of backrest, raising of seat and raising of backrest not by means of malleable cast or pressed steel handles or hand wheels.		X
Chair frames, Type I	Ends of arm rest tubing not closed by notching and welding where secured to seat frame.		X
	Reinforcements at corner connections not as specified.		X
	Lower end of legs of Type I chairs not closed and pierced to receive glides.		X
Chair frames, Type II	Seat frame not channel formed and reinforced.	X	
	Ends of arm rest tubing not closed by notching and welded where secured to seat frame.		X
	Seat frame not reinforced with steel brackets at point of attachment to spider arm.		X
	Back frame not one continuous piece.	X	
	Back frame not "U" shape when attached		X
	Cross brace not welded between legs of back frame.		X
	Leg of lower part of back frame not reinforced.		X
	Back frame not adjustable vertically and horizontally.	X	

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Classification of Defects (cont.)

Examine	Defect	Major	Minor
Chair frames Type II (cont.)	Aluminum pilot bearing plates not assembled into open top ends of back frame.	X	
Chair frames	Top ends of back frame tubing not closed by notching and welded into radius.	X	
Workmanship and construction	Any part missing, broken, split, fractured, cracked or punctured.	X	
	Any part malformed, dented, buckled, warped, bent out of shape, or otherwise damaged.	X	
Workmanship and	Any part misaligned, mispositioned, or in wrong location affecting serviceability.	X	
	Sharp edge or projection that may cause injury or damage to clothing.	X	
	Any part not properly or securely attached affecting safety.	X	
	Any part not properly or securely attached but not affecting safety.		X
Welds	Omitted where specified.	X	
	Not specified type.	X	
	Not fused.	X	
	Burnt through.	X	
	Cracked or fractured.	X	
	Flash or spatter not removed.		X
	Pitted.		X
	Not smooth.		X

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Classification of Defects (cont.)

Examine	Defect	Major	Minor
	Not complete.		X
Identification Marking	Missing.	X	

4.3.2.2 Dimensional examination. Inspection shall be made of chairs for compliance with the dimensions specified, including thickness of metal and film thickness of finish. Any dimension not within the tolerance specified will be classified as a defect. The inspection level and the acceptable quality level shall be as specified in 4.4.

4.3.3 Examination of preparation for delivery. An examination shall be made for compliance with packing, packaging, and marking provisions of section 5 of this specification. Defects will be scored as specified in the table below. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The acceptable quality level shall be as specified in 4.4.

Preparation for Delivery Defects.

Examine	Defect
Marking	Omitted; incorrect; illegible; or improper location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of taped seal, container flaps, loose strapping, inadequate cushioning, bulging, or distortion of containers.

4.4 Inspection levels and AQL's. The inspection levels and the acceptable quality levels (AQL's) expressed in defects per hundred units shall be as follows:

For examination in	Inspection level	AQL's Major	AQL's Total
4.3.2.1	II	4.0	10.0
4.3.2.2 (1 class)	II		4.0
4.3.3 (1 class)	S-2		4.0

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4.5 Tests. The following tests shall be performed as described herein on each style and class of chair being produced under contract. Tests shall be performed on the first article samples submitted for inspection and on a quarterly basis for the duration of the contract using three complete chairs for the testing. Tests may be additionally performed at the discretion of the Government (see 4.1). Any cracks or fractures, failure of welds, mechanical joints or mechanism, or any other failure that does not meet the requirements specified herein shall be cause for rejection of the style and class of chair being tested and no shipments of chairs shall be made until the problems causing rejection are resolved.

4.5.1 ANSI/BIFMA tests. Each type, style, and class of chair shall comply with the applicable requirements in ANSI/BIFMA X5.1-1985 and with the following deviations therefrom:

A) In section 12, change Figure 8 to show a drop height of 50 ± 2 mm above the uncompressed seat surface.

B) In section 12.3, delete test procedure (3) entirely and in test procedure (4), delete height of drop and bag weight entirely and substitute, "Height of drop shall be 50 ± 2 mm above the uncompressed seat surface. Bag weight shall be 56.7 ± 0.5 kg."

C) In section 18, perform both the soft tread caster test and the hard tread caster test on each chair.

4.5.2 Finish hardness. A film of the coating specified in 3.1.5 shall be applied to a sample of the anodized aluminum used in manufacture of the chair to a dry film thickness of 0.010 ± 0.001 mm. The film shall show no evidence of scratching when a drawing pencil of 4H hardness with a flat, square point (formed by sandpapering) held at an angle of 0.70 ± 0.04 rad is drawn with firm pressure across the surface in an opposite direction to the angle.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be as specified in the contract or purchase order (see 6.1).

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and should specify the following:

(a) Title, symbol, and date of this specification.
(b) Type, style, and class of chair required (see 1.2).

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(c) Color of upholstery required (see 3.1.8 and 3.1.9).
(d) Responsibility for inspection (see 4.1).
(e) Packaging, packing, and marking required (see 5.1).

Preparing Activity:
GSA-FSS/3FNE

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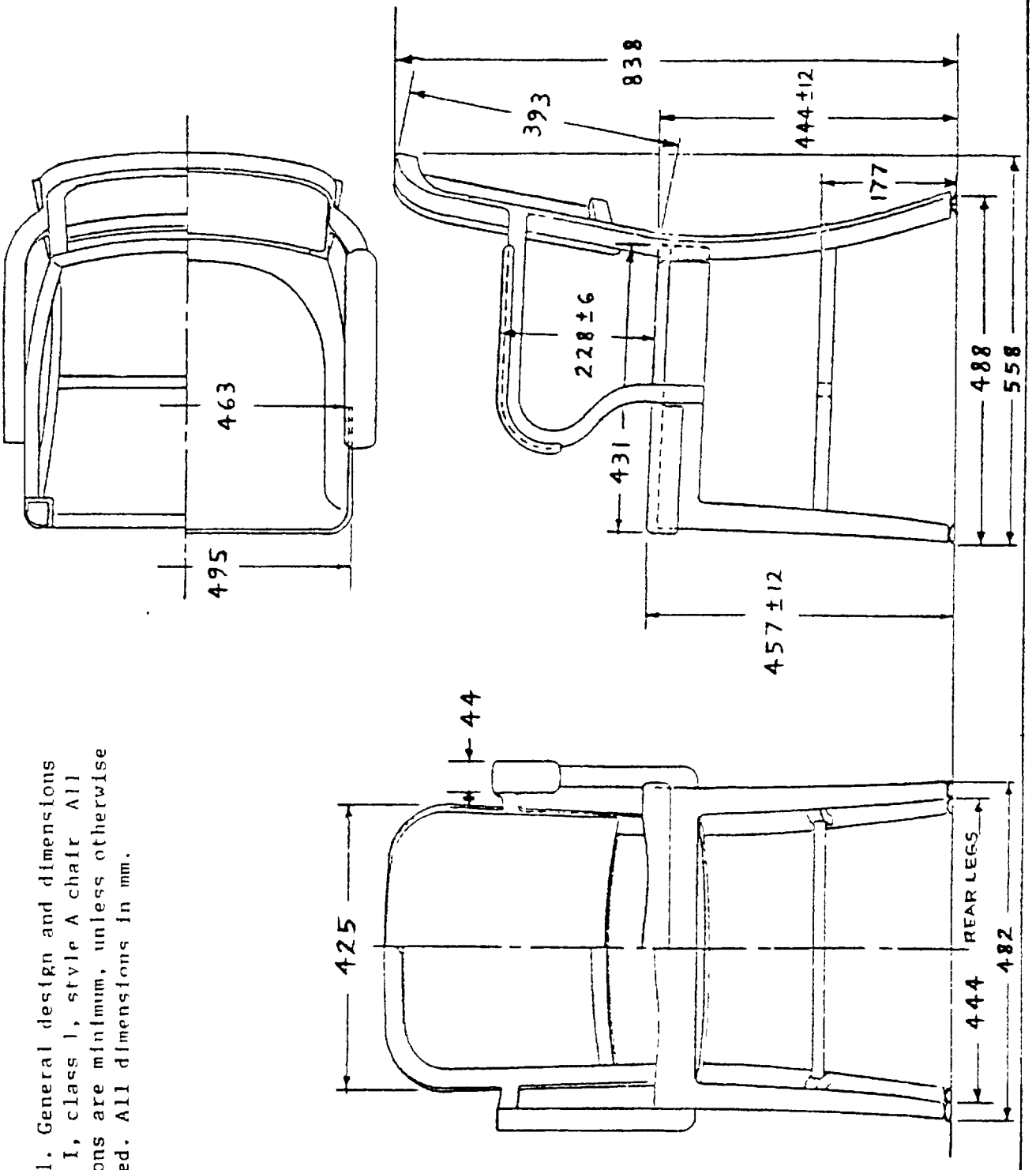


Figure 1. General design and dimensions of type I, class 1, style A chair. All dimensions are minimum, unless otherwise indicated. All dimensions in mm.

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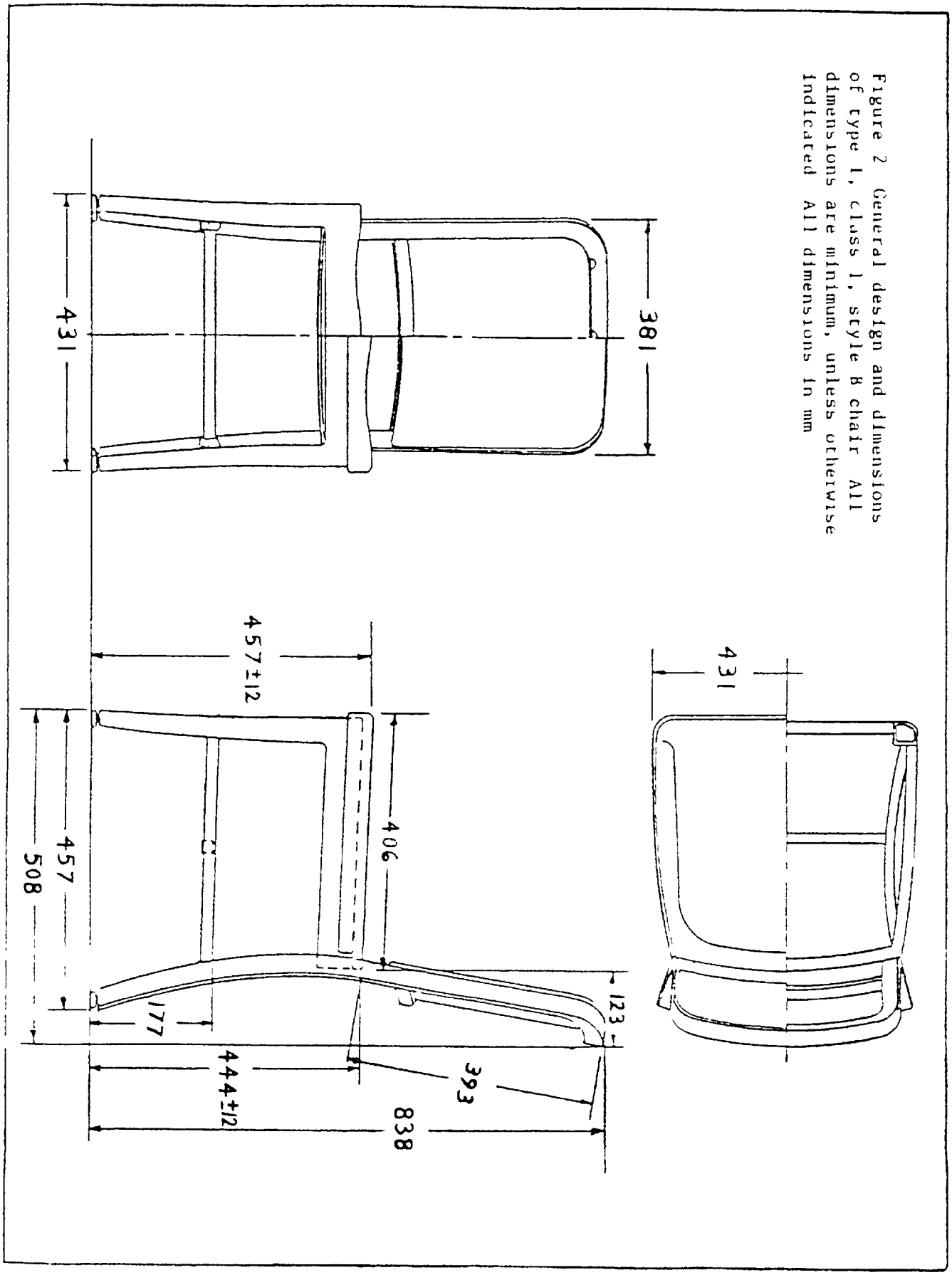


Figure 2 General design and dimensions of type I, class 1, style B chair All dimensions are minimum, unless otherwise indicated All dimensions in mm

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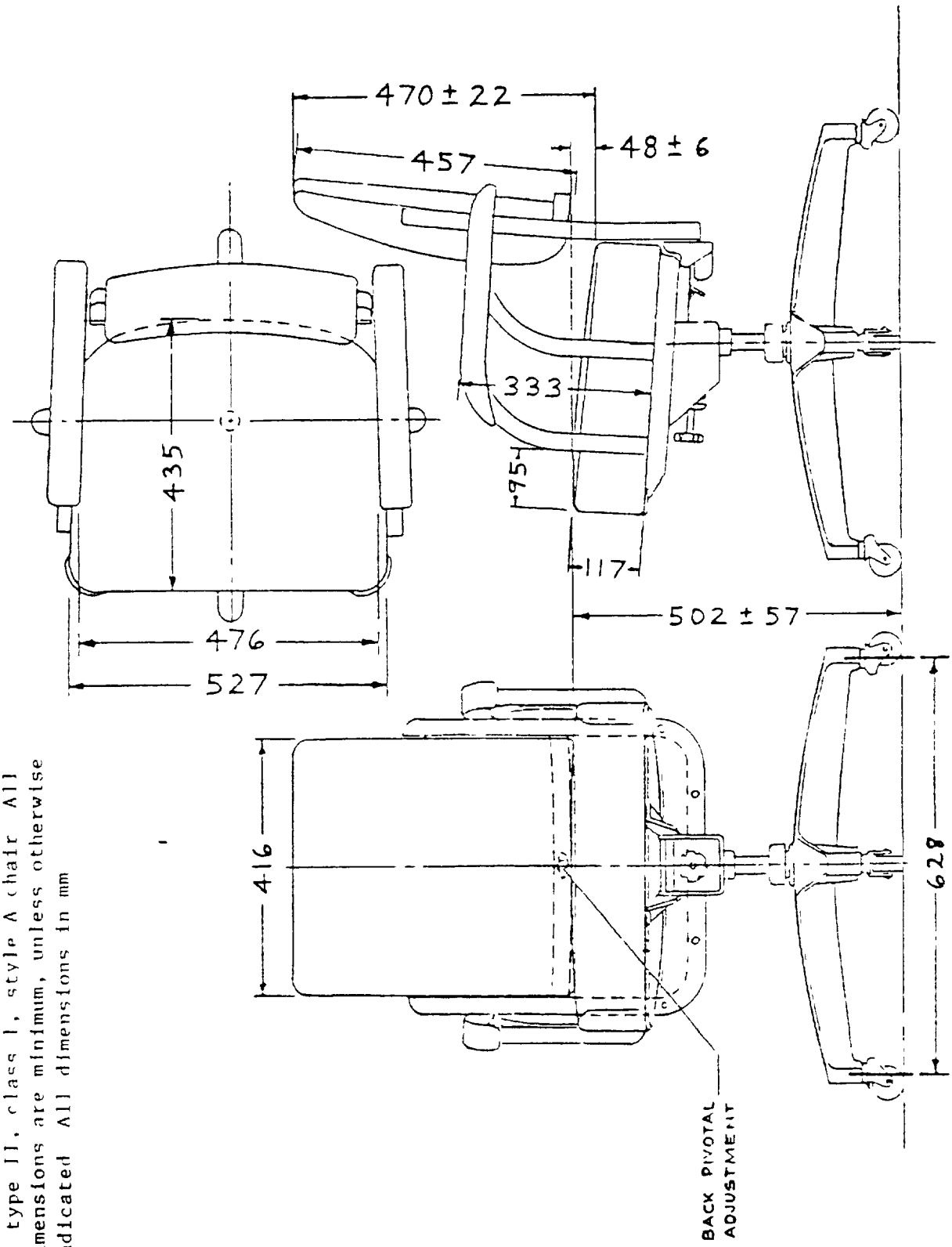


Figure 3 General design and dimensions of type II, class 1, style A chair All dimensions are minimum, unless otherwise indicated All dimensions in mm

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